

We consider distinct positive integers $a_1, \dots, a_k$ (with $k \geq 3$), product $P = \prod a_i$, and define $b_i = P/a_i + 1$. Let $g = \gcd(b_1, \dots, b_k)$. We want the maximum possible g under the condition $\sum a_i < 3 \cdot 10^6$.

Necessary conditions for g . Since $g \mid b_i$, we have $b_i = gm_i$. Then $P + a_i = a_i b_i = a_i gm_i$, so $g \mid (P + a_i)$. Hence $P \equiv -a_i \pmod{g}$ for all i ; consequently all a_i are congruent modulo g . Denote the common residue by r (we may take $1 \leq r \leq g-1$ because if $r = 0$ then $P \equiv 0 \pmod{g}$ but from later $r^{k-1} \equiv -1$ gives a contradiction for $g > 1$).

For each i , $b_i = \prod_{j \neq i} a_j + 1$. Since $a_j \equiv r \pmod{g}$, we have $\prod_{j \neq i} a_j \equiv r^{k-1} \pmod{g}$. Because $g \mid b_i$, this yields $r^{k-1} + 1 \equiv 0 \pmod{g}$, i.e.

$$r^{k-1} \equiv -1 \pmod{g}.$$

In particular, $\gcd(r, g) = 1$ (any common prime divisor would divide -1 modulo g).

Write $a_i = r + gt_i$ with integers t_i . Because $1 \leq r \leq g-1$ and $a_i > 0$, we must have $t_i \geq 0$. The distinctness of the a_i gives the t_i distinct. The sum condition becomes

$$\sum a_i = kr + g \sum t_i < 3 \cdot 10^6.$$

Since the t_i are distinct non-negative integers, $\sum t_i \geq 0 + 1 + \dots + (k-1) = k(k-1)/2$. Hence from (2)

$$kr + g \cdot \frac{k(k-1)}{2} < 3 \cdot 10^6.$$

In particular, for $k = 3$ we obtain

$$3r + 3g < 3 \cdot 10^6 \implies r + g < 10^6.$$

For $k \geq 4$, (3) gives $g \cdot k(k-1)/2 < 3 \cdot 10^6$, so $g < 6 \cdot 10^6/(k(k-1))$. For $k = 4$ this yields $g < 500\,000$; for $k = 5$, $g < 300\,000$; etc. Therefore any value of g larger than $500\,000$ can only be attained when $k = 3$. Since we will later exhibit an example with $g > 500\,000$, the maximum possible g must come from $k = 3$; we may thus restrict ourselves to $k = 3$ in the search for the maximum.

Reduction for $k = 3$. For $k = 3$ we have (1): $r^2 \equiv -1 \pmod{g}$, and (4): $r + g < 10^6$. Conversely, if integers r, g satisfy $r^2 \equiv -1 \pmod{g}$ and $r + g < 10^6$, then taking $a_1 = r$, $a_2 = r + g$, $a_3 = r + 2g$ yields distinct positive integers with sum $3(r + g) < 3 \cdot 10^6$, and because $a_i \equiv r \pmod{g}$ and $r^2 \equiv -1$, each b_i is divisible by g . Hence such a pair (r, g) produces an admissible set whose gcd is at least g .

For any admissible set with $k = 3$, the actual gcd g itself satisfies the two conditions (by the necessity argument). Therefore the greatest possible value of g across all choices (with any k) is exactly the largest integer D for which there exists an integer r with

$$r^2 \equiv -1 \pmod{D} \quad \text{and} \quad r + D < 10^6.$$

We now determine this maximum D .

Upper bound. Let D satisfy (5) for some r . Write $r^2 + 1 = tD$ with an integer $t \geq 1$. Since $D > r$ (the residue r is taken in $[1, D - 1]$), we have $t = (r^2 + 1)/D < (r^2 + 1)/r = r + 1/r$, so $t \leq r$. Condition $r + D < 10^6$ becomes

$$r + \frac{r^2 + 1}{t} < 10^6 \iff r^2 + tr + 1 < t \cdot 10^6.$$

Assume $D > 998285$. Then from $r + D < 10^6$ we obtain $r < 10^6 - D < 1715$, so $r \leq 1714$. Consequently

$$r^2 + 1 \leq 1714^2 + 1 = 2\,937\,797.$$

Since $D > 998285$, the quotient $t = (r^2 + 1)/D$ satisfies $t \leq 2\,937\,797/998286 < 3$; thus $t = 1$ or $t = 2$.

- If $t = 1$, then $D = r^2 + 1$ and $r + D = r^2 + r + 1 < 10^6$. The last inequality forces $r \leq 999$ (because $1000^2 + 1000 + 1 = 1\,001\,001 > 10^6$), so $D \leq 999^2 + 1 = 998\,002$. This contradicts $D > 998285$.
- If $t = 2$, then $D = (r^2 + 1)/2$ and $r + D = r + (r^2 + 1)/2 = (r + 1)^2/2 < 10^6$ gives $(r + 1)^2 < 2 \cdot 10^6$. Hence $r + 1 \leq 1414$ (since $1414^2 = 1\,999\,396 < 2 \cdot 10^6$ and $1415^2 > 2 \cdot 10^6$), so $r \leq 1413$. Then $D \leq (1413^2 + 1)/2 = 998\,285$, again contradicting $D > 998285$.

Therefore no integer $D > 998285$ can satisfy (5). Hence any admissible g (for any k) satisfies $g \leq 998285$.

Achievability. Take $k = 3$, $r = 1413$, $g = 998285$. Then $r^2 + 1 = 1\,996\,569 + 1 = 1\,996\,570 = 2 \cdot 998285$, so $r^2 \equiv -1 \pmod{g}$. Also $r + g = 1413 + 998285 = 999\,698 < 10^6$. Set

$$a_1 = 1413, \quad a_2 = 1413 + 998285 = 999\,698, \quad a_3 = 1413 + 2 \cdot 998285 = 1\,997\,983.$$

They are distinct positive integers and their sum is

$$1413 + 999\,698 + 1\,997\,983 = 2\,999\,094 < 3 \cdot 10^6.$$

Because $a_i \equiv r \pmod{g}$ and $r^2 \equiv -1 \pmod{g}$, we have

$$b_1 = a_2 a_3 + 1 \equiv r \cdot r + 1 = r^2 + 1 \equiv 0 \pmod{g},$$

and similarly $b_2, b_3 \equiv 0 \pmod{g}$. Hence g divides all b_i ; therefore the greatest common divisor of the b_i is at least g . But we have proved that no admissible set can have a common divisor larger than 998285; consequently the gcd for this set is exactly 998285.

Thus the value 998285 is attained as the largest integer dividing each b_i .

Conclusion. The maximum possible value of g across all choices of distinct positive integers $a_1, \dots, a_k$ ($k \geq 3$) with product P and sum $< 3 \cdot 10^6$ is

$$\boxed{998285}.$$